4.7 Exercises

1. **Linear State Observers**
   - Considering the model from equation (4.1) write a simulation and implement a state observer to estimate the system's velocity from position measurements and compare the results, qualitatively and quantitatively, with the position difference estimator and the acceleration integration estimator for different values for the measurement noise variance and the input uncertainty variance, but disregarding (setting to zero) the input uncertainty bias.
   - Repeat the procedure from the previous item, but now considering the effects of input uncertainty bias. Adapt the state observer design to compensate for the input uncertainty bias.
   - Implement and simulate a linear Kalman filter to estimate the system's velocity from position measurements and compare the results, qualitatively and quantitatively, with the linear state observer designed in the first item for different values for the measurement noise variance and the input uncertainty variance, but disregarding (setting to zero) the input uncertainty bias.
2. **Nonlinear Kalman Filters**
   (a) Considering the model from equation (4.130), write a simulation and implement an extended Kalman Filter and an Unscented Kalman filter to estimate the system's state from position (GNSS), acceleration and angular velocity measurements (IMU) and compare the results, qualitatively and quantitatively, for different values for the measurement noise variance, the input uncertainty variance and GNSS update period (as a multiple of the time step, considering that the time step is equal to the IMU sampling period), but disregarding (setting to zero) the input uncertainty bias.
   (b) Repeat the procedure from Item a. but now considering the effects of input uncertainty bias. Adapt the Kalman Filters to compensate for the input uncertainty bias.
3. **Simulation**
   (a) Implement a Kalman Filter, using a method of choice, to estimate the states of an autonomous vehicle from the Carla simulator using measurements from the GPU and IMU provided by the simulator, including measurement noise and angular velocity measurement bias. (Tip: remember to transform latitude, longitude and altitude in meters before using the GPS measurements.